



Linking firm performance with innovation culture: An algorithmic approach towards theory building

Wanqing Li^{a,b}, Jiang Yu^{a,b,*}, Feng Chen^{a,b}

^a The Institutes of Science and Development, Chinese Academy of Sciences, Beijing 100190, China

^b The School of Public Policy and Management, University of Chinese Academy of Sciences, Beijing 100049, China

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ABSTRACT

To address the growing demand for theory development in computational social science research, this study employs an algorithm-driven approach to formulate a comprehensive six-step theory generation process. By applying this original research method, a new theoretical model—the “Dynamic Resource-Culture Synergy Theory” is proposed which enhances the explanatory power regarding how firms maintain competitiveness in rapidly changing environments by emphasizing the pivotal role of culture in resource integration and innovation processes. Drawing on empirical data from 887 Chinese high-tech manufacturing firms, our analysis identifies key drivers of organizational performance, with a particular focus on the role of organizational culture, especially innovation culture, as a mediating force. Utilizing the GWO-SVM technique, we gain a nuanced understanding of how different cultural traits interact with innovation and leverage, uncovering how the initial enhancement of innovation culture positively impacts performance metrics such as ROA. The findings confirm that innovation, facilitated by organizational culture, significantly enhances performance outcomes. Furthermore, this study considers factors such as leverage and the proportion of technical personnel, investigating their moderating effects on the relationship between innovation culture and firm performance. This study not only deepens the understanding of how innovation and culture interact to influence firm performance but also provides significant theoretical and practical contributions to the study of the dynamics of high-tech manufacturing enterprises.

1. Introduction

The importance of theory in a research endeavor has been widely recognized and theory building plays a crucial role in most research disciplines (Svejvig, 2021). In management research, the mutual strengthening between good management practices and good management theories contributes to the prosperity and development of both real business world and academic circle (Ghoshal, 2005). In general, qualitative research is one of the most frequently-used approaches in generating theories. However, generating theories from data analysis is relatively less adopted (Shrestha et al., 2021). Nowadays, various new approaches like big data and artificial intelligence (AI) provide more options for management scholars to undertake the theory-generation tasks in new ways (Arora et al., 2016; Cao, 2022).

In the process of forming new theories, two theoretical construction logics of the mirror of each other, inductive theory (generalize theories from cases and propose new ones) and deductive theory (emphasize the

test of theories through data) have different advantages and shortcomings (Eisenhardt and Graebner, 2007). Specifically, the inductive method is constrained in theoretical validation while the deductive method lacks conceptual precision and logical theoretical argument (Davis et al., 2007). In order to demonstrate the advantages of both methods, one important task is to find the “sweet spot” between the inductive-based theory-creating research and the deductive-based theory-testing research.

Scholars pointed out that traditional theory-driven research often face with challenges, including data overfitting (Kar and Dwivedi, 2020), failures to consider the relationships among factors within the context (Grover, 2020). Nowadays, more and more advanced analytical techniques and tools including various data-driven algorithms are introduced to theory building efforts (Shrestha et al., 2021; Bamel et al., 2022; Skare and Soriano, 2021). Take AI as an example, AI techniques provide high efficiency in analyzing relevant data through approach that is similar to the logic of grounded theory. It shows its full potential as a

* Corresponding author at: The Institutes of Science and Development, Chinese Academy of Sciences, Beijing 100190, China.

E-mail addresses: Vinnie.LiWQ@126.com (W. Li), yujiang@casisd.cn (J. Yu), chenfeng20@ucas.ac.cn (F. Chen).

novel methodology for theory building in management research (Choudhury et al., 2021). However, how the tools can be applied in the theorizing process (Shepherd and Suddaby, 2017) requires further investigation, especially the clarification of the procedure of theory construction with new data-driven algorithms in management field.

In the epoch of Big Data, it becomes paramount for analytical mechanisms to dynamically elucidate optimal cause-effect correlations to stakeholders, seamlessly integrating the retrospective insights gleaned from descriptive analytics with the prospective foresight proffered by predictive analytics (Deshpande et al., 2019; Wamba-Taguimdje et al., 2020). Within this framework, the salient role of prescriptive analytics becomes evident. Prescriptive analytics elucidates the most favorable outcomes under specified conditions and recommends strategic pathways to harness forthcoming opportunities or to offset impending threats. Furthermore, in light of business limitations, it delineates the paramount strategies and offers profound understanding of prospective results (Deshpande et al., 2019; Charles et al., 2022). Across diverse domains such as manufacturing, marketing, human resources, business strategy, and health, prescriptive analytics has proven instrumental. Amidst the rapid developments in AI and big data technologies, prescriptive analytics has found its ground to offer innovative solutions (Mariani and Wamba, 2020). Based on the comprehensive literature review, six primary categories emerge within the realm of prescriptive analytics methodologies: Probabilistic Models, Machine Learning/Data Mining, Mathematical Programming, Evolutionary Computation, Simulation, and Logic-based Models (Lepeniotti et al., 2020).

To address the proposed research question, we take an AI-based perspective and employ Machine Learning (ML) algorithms to illustrate the algorithm-driven theory generation process. The ML techniques is selected for the following reasons. ML techniques are used to analyze high-throughput data to gain useful insights, classifications, predictions, and evidence-based decisions, which has led to the widespread use of ML techniques in many areas including information science, mathematics, management and among others (Xu et al., 2021). Miric et al. (2023) summarize the application of ML algorithms in strategy and management researches as: (1) observations classification to create categorical variables (Lu and Chesbrough, 2022), (2) implicit pattern identification (Choudhury et al., 2021), (3) developing existing econometric methods (Vanneste and Gulati, 2022), (4) processing textual and image data, and (5) recognition of input variables (Nauhaus et al., 2021). ML methods can produce observations from data for the theory building process (inductive or abductive), which play a central role in management and organizational research (Choudhury et al., 2021; Shrestha et al., 2021). However, researches on paradigm shifts in ML-based reasoning for theory building are not as broad as researches on the application of ML in management and organizational field (von Krogh, 2018). As a result, an in-depth exploration of ML-based theory building process and its application in the field of organization management is needed.

The reason for selecting firm performance as the research theme lies in the fact that organizational performance has consistently been a focal point in the field of enterprise management studies. Scholars have extensively contributed viewpoints and empirical analyses to this topic, thereby furnishing our ML algorithm with a substantial literature database. Simultaneously, we have observed that a portion of previous research on firm performance has employed regression analysis (Menter et al., 2023), GMM (Saeed et al., 2023), SEM (Chatterjee et al., 2023), PCA (Usai et al., 2021). However, the use of ML algorithms is not very common in studies related to firm performance (Nyagadza, 2022; Martínez et al., 2022). Simultaneously, the lack of uniformity in research results is influenced by variations in approaches to measuring firm performance, research design, sampling practices, and addressing endogeneity issues. This paper employs a ML algorithm model, which not only effectively address issues related to sample selection and research design, but also examines and extends this model through a comparison with traditional econometric methods, highlighting the

advantages of the ML algorithm-driven theoretical generation model. More importantly, our performance analysis incorporates organizational culture, a factor traditionally recognized as a key element in business success (Cefis and Marsili, 2006). Due to the complexity of surveying and the resource-intensive nature of organizational culture studies, their richness has not met expectations. In an innovative approach, we use ML algorithms to quantify organizational culture, providing new perspectives for ML algorithm applications in organizational theory research.

In this study, we employ a novel, multi-staged algorithm-driven approach to investigate the complex interplay between innovation, organizational culture, and firm performance, specifically in the high-tech manufacturing sector in China. Our methodology begins with the establishment of a theoretical framework using Latent Dirichlet Allocation (LDA) modeling. This is followed by an in-depth analysis of organizational culture using advanced natural language processing techniques. Subsequently, we quantify these theoretical constructs and explore their associations using Support Vector Machine (SVM) analysis. We further involve empirical validation of our findings through regression analysis and comparison with traditional econometric methods. Finally, we refine the theoretical constructs using the GPT-4 language model, exploring and presenting the theory through diverse narrative styles to enhance clarity and comprehension. This comprehensive approach allows us to deepen the understanding of how innovation and culture interact to influence firm performance, offering significant theoretical and practical insights into the dynamics of high-tech manufacturing businesses.

This study introduces the Dynamic Resource-Culture Synergy Theory, which integrates the Resource-Based View (RBV) and organizational culture theory, offering a novel perspective on how firms achieve and sustain competitive advantage in rapidly changing environments. Unlike traditional static models, this theory highlights the dynamic moderating role of organizational culture in resource integration and innovation processes, providing deeper insights into how culture functions as an intangible resource that influences a firm's innovation activities and overall competitiveness. Furthermore, the study leverages data-driven empirical methods to enhance the theory's predictive capability and applicability across various organizational and industry contexts, laying the groundwork for further theoretical development.

2. Literature review

In this section, we first examine the dynamic relationship between corporate innovation and performance in Section 2.1, highlighting the significant impact of innovation strategies on business outcomes. Section 2.2 then explores the role of organizational culture in influencing firm performance, delving into its multifaceted contributions. In Section 2.3, we focus on the transformative role of AI and ML in contemporary management research, particularly in their application to theory generation and data analysis.

2.1. Innovation and performance in the corporate sector

A considerable body of scholarly work has delved into the dynamics between corporate innovation and performance, consistently unveiling a trend towards favorable outcomes. As highlighted by Walker (2004) among others, innovation emerges as a pivotal factor in elevating firm performance, enhancing market stance and competitive edge, which culminates in superior organizational achievement (Gunday et al., 2011). Further, elevated innovation levels within firms correlate with marked improvements in task efficiency, organizational effectiveness, and the establishment of lasting competitive benefits (Khan et al., 2018). The realm of innovation capabilities has been recognized for its salient positive contributions to product quality, operational excellence, and financial robustness (Kafetzopoulos and Psomas, 2015). This positive trend is accentuated by a significant nexus linking innovation capabilities with both market success and financial prosperity of companies

(Rajapathirana and Hui, 2017). From the perspective of service-dominant logic, Yu et al. (2019) emphasizes collaborative processes and resource integration, illustrating the relationship between innovation and performance. Moreover, Liu et al. (2022) underscores the pivotal role of strategic resource configuration within an innovation ecosystem, demonstrating how it can substantially enhance firm performance by driving technological advancements and facilitating market-oriented growth. From a comprehensive standpoint, the spectrum of innovation performance, covering product, process, marketing, and organizational dimensions, plays a crucial role in amplifying a range of business metrics, encompassing market visibility, non-financial, and financial performance.

Nonetheless, it is crucial to acknowledge divergent findings in some research streams, which point to potential negative or non-correlations between innovation and business performance (Subramanian and Nilakanta, 1996). Furthermore, while existing literature extensively documents the positive impact of innovation on performance, it often lacks a nuanced exploration into how innovation manifests in various industries and scales of businesses, as well as the interplay between innovative strategies and organizational culture. Notably, current research tends to emphasize the aggregate effects of innovation, falling short in examining its function within diverse organizational contexts and its interaction with internal management and strategic decision-making processes. Considering this, the present study seeks to bridge these gaps by delving into the specific applications of innovative strategies across different organizational and market settings, and their subsequent influence on firm performance.

2.2. Organizational culture and its impact on innovation and performance

Organizational culture is widely acknowledged as a critical determinant of business success (Hogan and Coote, 2014), consistently representing a focal point in the field of management studies. The diversity of perspectives individuals adopt makes it difficult to reach a universally accepted definition (Rollinson, 2002). Jacques (1952) was among the pioneers in positing that organizational culture embodies habitual and traditional ways of conducting tasks, shared to varying extents by all members. New members must learn and embrace these practices to be accepted into the company. Organizational culture can also be defined as the shared norms, values, social ideals, or beliefs among the members of an organization (Martins and Terblanche, 2003). Undoubtedly, valuable, rare, and non-imitable organizational culture is recognized as one of the resources contributing to an organization's sustainable profitability (Barney, 1986) and supports firm performance.

Simultaneously, advances in organizational economics conceptualize organizational culture as shared beliefs and examine its link to performance through the processes such as selection, self-sorting, and manager-directed learning (Tan, 2019). Mohammadi et al. (2023) found that organizational culture, encompassing elements such as goal transparency, coordination and integration, dispute resolution, and employee participation, significantly influences corporate financial innovation performance.

In the exploration of the relationship between organizational culture and performance, scholars have also introduced organizational innovation into their studies. Weick and Quinn (1999) found that components of organizational culture influence performance, particularly in initiating new challenges and substantial transformations, fostering the development of an innovative organization (Carmeli and Gefen, 2005). Boubakri et al. (2021) asserts that organizational culture significantly influences corporate innovation, with organizational culture being a vital antecedent of innovation (Dani and Gandhi, 2021). Simultaneously, organizational learning culture has a moderately indirect positive impact on innovation through the lens of innovation culture (Škerlavaj et al., 2010), with this mediating role being evident in the impact of strategic innovation dimensions on product, process, and

market innovation.

Despite the richness of scholarly research on organizational culture, organizational performance, and innovation, there remains a lack of research on the specific pathway through which innovation in organizational culture enhances firm performance. Moreover, the measurement methods for organizational culture largely rely on scale statistics and regression analyses based on specific populations. In the backdrop of the era of data intelligence, there is an attempt to explore more comprehensive and nuanced AI methods, aiming not only to enhance efficiency but also to improve the precision of results.

2.3. The role of AI and ML in contemporary reasoning methods

To fully grasp the role of algorithm-supported induction in the construction of theories, it is imperative to delve into the diverse forms of reasoning. Within the domain of scientific reasoning and inquiry, three fundamental types of logical reasoning stand paramount: Abduction, Deduction, and Induction, are considered the essence of logical thinking. Abductive reasoning entails formulating hypotheses about the most plausible causes for observed phenomena, anchored in the bedrock of existing theoretical frameworks. This reasoning mode effectively utilizes established knowledge to derive conceivable explanations. In contrast, deductive reasoning employs overarching principles to predict specific outcomes, upholding logical integrity and factual accuracy. Inductive reasoning, meanwhile, garners generalized principles from iterative observations, thus enhancing and expanding the corpus of existing knowledge (Upmeyer et al., 2021). Scholars adeptly outlines the intricate interplay among these forms of reasoning: Deduction establishes what necessarily is, Induction unveils what is actively operational, and Abduction proffers insights into what could potentially be. Emblematic of abductive reasoning, grounded theory represents a data-oriented, theory-formulation methodology that adeptly navigates the continuum between inductive empiricism and deductive theoretical constructs, culminating in the derivation of theories that are deeply rooted in empirical data (Murphy et al., 2017).

The recent surge in AI and ML has significantly influenced theoretical advancements across various disciplines. ML, being inherently neutral regarding existing theories, permits the data itself to reveal underlying patterns. This trait makes ML an exceptional instrument for exploratory investigations. ML's ability to discern features in data sets that align with specific outcomes has proven beneficial in enhancing causal models, thereby complementing traditional methods in economics. For instance, the role of ML in pattern detection serves as an invaluable adjunct to conventional econometric hypothesis testing, especially in inductive or abductive research, by uncovering complex data patterns without predetermined structures, thus facilitating the process of theory formation.

In the strategic realm, AI's potential to transform industrial processes and exert a multifaceted impact on organizations and societies is undeniable. Although encompassing an array of statistical prediction techniques, AI's deployment in entrepreneurship research exemplifies its ability to foster theoretical growth and bridge the gap between theory and practice (Crown, 2019). Researchers have also harnessed ML to refine existing econometric techniques, using algorithms for automatic control variable selection and instrumental variable prediction, leading to more robust causal inferences from better-matched samples or varied experimental treatment effects (Miric et al., 2023).

Furthermore, ML's prowess in managing large data sets and conducting intricate analyses underpins the training of algorithms, crucial for precise predictions and pattern recognition essential to theory development (Kumar et al., 2020). Such patterns identified by ML are instrumental in exploratory induction or abductive studies, as well as in post-hoc regression analysis, thereby drawing quantitative empirical researchers closer to the principles of grounded theory, where the identification of patterns in data is fundamental (Shrestha et al., 2021).

Through the synthesis of the literature, it becomes evident that a

primary challenge in contemporary research is the interpretation of increasingly complex and voluminous data. This is a task that traditional forms of reasoning — abduction, deduction, and induction — find particularly demanding to address in a comprehensive manner. Such a scenario underscores the imperative need for a more nuanced and efficient approach to data interpretation. This is especially true in the field of organizational management, where the integration of these diverse reasoning methods is crucial in managing extensive and heterogeneous data sets. Additionally, in disciplines such as economics and management research, the development of robust causal models is frequently impeded by the inherent limitations of conventional methodologies when grappling with complex and non-linear data patterns. A further notable challenge lies in the reconciliation of theoretical constructs with practical applications, a task that is increasingly pressing in the dynamically evolving realm of organizational management.

In this context, algorithm-supported induction, harnessing the power of AI and ML, emerges as a formidable solution to these multifaceted challenges. It significantly augments data processing capabilities, facilitating the analysis of expansive data sets and the unveiling of intricate patterns that might elude traditional analytical methods. The application of ML in pattern detection and feature identification within these data sets markedly enhances causal modeling, representing a substantial advancement over conventional hypothesis testing approaches. Moreover, algorithm-supported induction adeptly narrows the chasm between theoretical frameworks and practical applications, particularly in the sphere of organizational management research.

3. Theoretical development and research hypothesis

3.1. Resource-based view

The RBV posits that when a firm possesses resources that are valuable, rare, inimitable, and non-substitutable (VRIN), it can attain sustainable competitive advantages by implementing value creation strategies that are difficult for competing firms to replicate (Barney, 1986). Company resources encompass all assets, capabilities, organizational processes, attributes, information, knowledge, etc., under the control of the firm (Daft, 1983). The value and rarity of resources enable firms to generate new economic value, while their inimitability and non-substitutability function act as isolating mechanisms that secure rents associated with those resources (Barney, 1991).

RBV contributes to explaining why some companies outperform others in the same industry. The accumulation of VRIN resources to enhance competitive advantage has become a fundamental academic and strategic concept in the field of management. For instance, Ozdemir et al. (2023) theoretically and empirically evaluate how stakeholders with different interests and motivations influence the development of collaborative innovation based on RBV theory. Zhang et al. (2022) examine the role of dynamic capabilities within the RBV framework and explore the relationship between different resources, dynamic capabilities, and firm performance. Zhang et al. (2023) explore the role of innovation in sustainable performance within the context of a circular economy from an RBV perspective, further enriching the application of RBV in management studies (Weinzimmer et al., 2023).

In addition, organizational culture, organizational capital structure, and strategy, as crucial intangible assets for enterprises, serve as sources of organizational innovation and act as driving forces for organizational change. These elements enable the organization to develop stronger dynamic capabilities and ultimately form its core competencies. Organizational culture is often regarded as a key factor for business success (Hogan and Coote, 2014). It not only shapes employees' values, attitudes, and behaviors but also attracts high-quality talent, leading to synergistic effects for the company's development. The relationship between organizational culture, organizational innovation, and performance has been extensively studied. However, how organizational culture translates its impact from organizational innovation to

organizational performance remains an under-researched question, and key factors such as organizational capital structure have not been fully considered.

3.2. Theoretical hypotheses

Innovation is a critical factor for companies to gain a competitive advantage and serves as a key driver of economic development, playing a crucial role in national and organizational competitiveness (Cefis and Marsili, 2006). Innovation motivates companies to introduce new products, gradually becoming pioneers in the market. Many factors have been proven to be determinants supporting an organizational innovation orientation, with organizational culture often recognized as a significant contributor to innovation (Griffith and Rubera, 2014). Existing literature has analyzed the interactive relationship between organizational culture and innovation. For instance, participants exposed to diverse cultures and possessing more diverse cultural knowledge generate more innovative ideas and solutions (Leung et al., 2008). Empirical analysis based on Spanish companies by Bendak et al. (2020) found that a democratic culture promotes innovation strategies, while a hierarchical culture promotes imitative culture, emphasizing the significance of organizational culture as a determinant of corporate innovation strategies. Some scholars position culture as the environment for innovation (Büschgens et al., 2013b). Organizational culture, through innovation-supportive policies, procedures, and management values, significantly impacts innovation. It has been identified as a robust predictor of open innovation, where a highly integrated culture provides a conducive environment for open innovation development (Naqshbandi and Tabche, 2018). Simultaneously, innovative companies often encourage employees to experiment and explore new methods, fostering an environment tolerant of failure and conducive to continuous learning. This further stimulates employees' creative thinking and provides an open, positive communication environment for the team. Hence, it can be boldly predicted that innovation has a significant impact on organizational culture.

H1. Innovation has a significant impact on organizational culture

According to the RBV theory, valuable, rare, and not easily imitable organizational culture is one of the resources that organizations can leverage for sustained profitability (Barney, 1986). Schein (1992) posits that organizational culture is a largely invisible yet powerful social force. Empirical evidence suggests that organizational culture significantly influences market-oriented behavior, market performance, and financial performance (Homburg & Pflesser, 2000). Weick and Quinn (1999) found that components of organizational culture impact performance, especially in initiating new challenges and substantial changes to foster the development of an innovative organization (Carmeli and Gefen, 2005). Mohammadi et al. (2023) discovered that elements of organizational culture such as goal transparency, coordination and integration, dispute resolution, and employee participation significantly influence financial performance. It can be broadly speculated that organizational culture contributes to the enhancement of organizational performance.

H2. Organizational culture has a positive impact on performance

Organizational culture is considered a source of sustained competitive advantage (Barney, 1991) and a key factor in organizational effectiveness (Denison, 1990). Büschgens et al. (2013b) position organizational culture as an environment for innovation. In such an environment, companies with a culture that encourages innovation can inspire employee enthusiasm and proactivity, encourage knowledge sharing and learning among employees, promote collaboration between teams, facilitate the rapid dissemination of new knowledge and skills, enhance project success rates, and consequently improve overall organizational performance. Additionally, a positive organizational culture background can reduce employees' concerns about innovation and fears of failure. Employees are more willing to experiment with new ideas and concepts, reducing innovation risks, increasing customer satisfaction,

better adapting to market changes, and maintaining a competitive advantage. Innovation, within the context of organizational culture, can better harness the inherent vitality of innovation itself, making the organization more adaptive and creative, ultimately contributing to improved organizational performance and long-term sustainable development. We hypothesize that innovation will be more effectively transformed into organizational performance through the influence of organizational culture.

H3. Innovation enhances organizational performance through organizational culture as a mediator

Relevant studies indicate that the asset-liability ratio significantly influences strategic investment, technological research and development, and the construction of innovative industries, all of which are crucial to the direction and prospects of a company's development (Miotto and Parente, 2015). Jiang et al. (2022) found that companies with higher leverage have a more urgent need for funds, resulting in a higher marginal utility of funds for these companies. According to marginal utility theory, this lead to a more significant improvement in their innovation performance. Studies using the asset-liability ratio as a measure of financial leverage have reached similar conclusions, showing that under certain conditions, companies with higher leverage often exhibit better innovation performance. However, this positive effect may not be universal; excessive leverage could lead to financial constraints that impede the effective allocation of resources, thereby negatively impacting outcomes. Hence, financial leverage, measured by the asset-liability ratio, plays a crucial role in influencing firm performance, potentially serving as a double-edged sword. Therefore, we introduce this variable into the study to examine its moderating role, considering its various potential impacts.

H4. The asset-liability ratio plays a moderating role on innovation

and performance

4. Algorithm-driven theory generation research processes in management

In this research, we implement a pioneering approach utilizing ML to forge management theory, with a focus on the high-tech manufacturing sector. Our methodology unfolds through six distinct stages, initiating with the establishment of theoretical scope and progressing to the extraction of innovation culture. Subsequently, we quantify these theoretical scopes, delve into associations using SVM analysis, and carry out empirical validation of our findings. The final stage involves the development of theoretical constructs. Throughout this process, advanced AI tools, notably GPT-4, play an instrumental role in deepening our grasp and articulation of intricate theoretical concepts. This novel strategy signifies a significant stride in the integration of AI within the domain of management theory generation. To provide a more cohesive understanding, we have also crafted a visual representation to complement the textual explanation. Fig. 1 succinctly captures the essence of each stage, offering a clear, visual pathway of our methodology.

Module 1: Theoretical framework establishment

This module establishes the foundational theoretical framework by leveraging advanced machine learning models such as LDA and BERT. It synthesizes insights from existing literature to construct a conceptual model that intricately links organizational culture, innovation, and performance, thereby providing a robust basis for the ensuing empirical analysis.

Stage 1. Forming theoretical scope

Our research, aimed at dissecting the complex dynamics within the

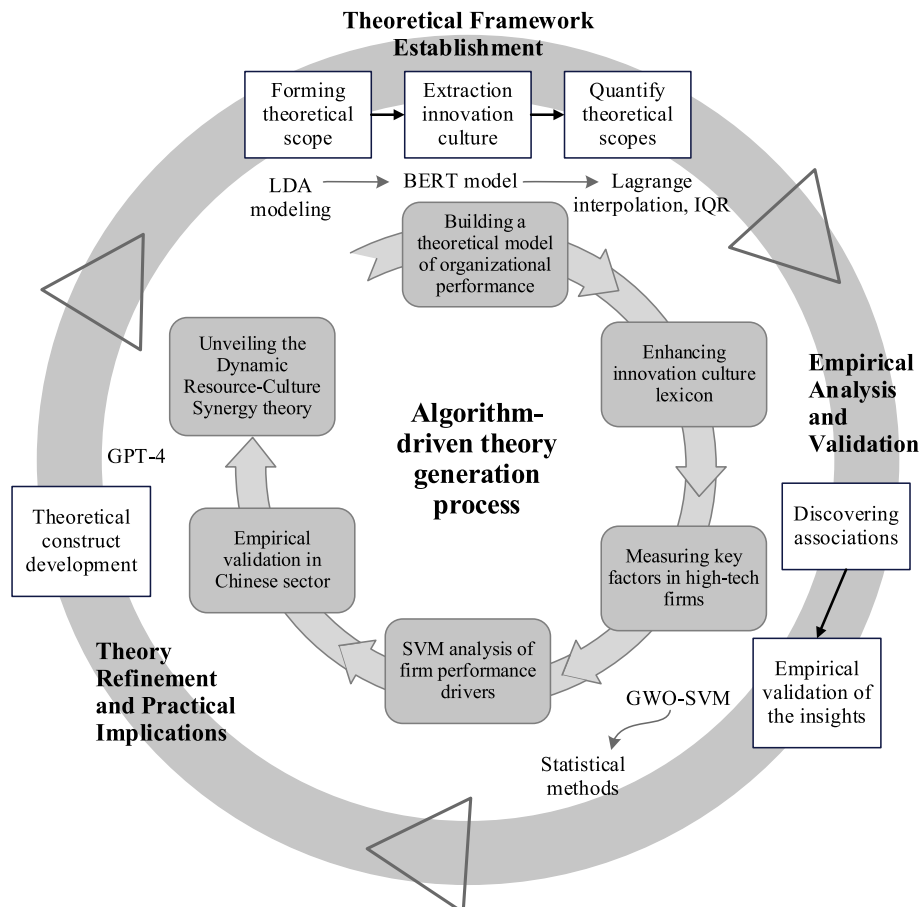


Fig. 1. Main stages of the methodology.

high-tech manufacturing sector, leveraged the LDA modeling algorithm to construct a multifaceted theoretical model encompassing organizational performance, innovation, and culture. This stage was pivotal for establishing a foundational framework, essential for decrypting the intricacies of organizational dynamics. Key to this theory generation phase was the integration and critical assessment of existing scholarly literature, pinpointing the principal elements and constructs impacting organizational performance. This crucial step not only paved the way for the ensuing empirical analysis but also anchored the theoretical model in scholarly discourse, thereby harmonizing theoretical concepts with empirical data.

The utilization of the LDA algorithm, an advanced ML tool, was central to our approach. It adeptly processed a vast array of textual data to uncover latent thematic structures, transforming complex literature into coherent thematic narratives. This method eclipsed traditional manual review methods by offering a more unbiased and comprehensive exploration of the research domain. The LDA algorithm's efficacy in identifying nuanced themes within the literature, guided by coherence scores, ensured the alignment of algorithmic precision with thematic clarity.

Our methodology's procedural steps involved compiling an extensive corpus from established databases, concentrating on key terms such as "performance", "innovation" and "culture". This led to the creation of a rich corpus consisting of 3814 articles. Rigorous data preprocessing was conducted to transform these articles into a format amenable for detailed analysis. This included segmenting the data into individual words and eliminating irrelevant content, thus refining the data quality for LDA analysis. The LDA algorithm subsequently identified distinct themes within the literature, each theme representing a critical aspect of organization culture, innovation and performance.

The insights derived from the LDA analysis facilitated the development of a comprehensive conceptual model. This model harmoniously integrated the identified themes, offering a structured depiction of the crucial elements shaping organizational performance. It represented the complex interplay of factors driving success in organizational landscape. Our adoption of the LDA-based approach substantially outperformed conventional research methodologies. It provided an objective viewpoint over a wide range of scholarly works, circumventing the typical biases of manual literature reviews and the potential overlook of pivotal studies, connected how the present study discovers themes and trends in a large collection of documents (Huang et al. 2022, Hannigan et al., 2019). This approach enabled a rapid yet thorough examination and analysis of a significant volume of literature, guaranteeing a broad and profound encapsulation of the research field. Moreover, it established a carefully curated, data-driven foundation, pivotal for a deep dive into the complex nature of organizational performance. This foundation crucially steered the later stages of our research, contributing to a more layered and robust understanding of the factors influencing performance in high-tech manufacturing enterprises.

Stage 2. Extraction innovation culture

In our exploration of organizational performance within the high-tech manufacturing sector, the extraction of the organizational culture variable stands as a cornerstone. Utilizing the advanced BERT (Bidirectional Encoder Representations from Transformers) model, we delved into the textual data of management discussions from annual reports. This intricate process was crucial for unearthing key aspects related to innovation culture, an essential element in our theoretical framework.

The BERT model, renowned for its deep contextual understanding, was pivotal in this endeavor. Through its nuanced analysis of text, it extracted significant insights about the underpinnings of organizational culture. The process began with diligent data preprocessing, followed by the extraction of embedding for keywords synonymous with innovation culture. These embedding were vital in identifying the most relevant terms related to innovation culture within the textual corpus. Drawing inspiration from the methodology outlined in Ref. (Li et al., 2020), we

utilized a set of seed words such as innovation, create, creative, efficient, innovate, innovative, and passionate. The initial lexicon, formed by selecting the top 100 words with the highest similarity scores, underwent meticulous manual refinement to ensure contextual relevance and accuracy.

This approach transcends traditional qualitative analyses, offering a more detailed and comprehensive lens for understanding organizational culture. Its objective and replicable nature significantly minimizes biases and human errors. Crucially, this step provides a quantifiable measure of innovation culture. This quantitative representation augments our theoretical model, facilitating a deeper understanding of the interplay between innovation culture and organizational performance, particularly within the high-tech manufacturing industry. This innovative measurement strategy not only enhances the depth of our theoretical model but also underscores the potential of advanced language processing methods in elevating theoretical research in management studies.

Stage 3. Quantify theoretical scopes

This pivotal stage of our methodology centers on empirically measuring critical elements within China high-tech manufacturing enterprises. It signifies a vital shift from formulating theoretical scopes to executing concrete data analysis, underpinning our theory generation process. During this phase, we meticulously chose high-tech manufacturing firms listed in China before 2016, ensuring a comprehensive and reliable dataset from various authoritative sources. Our focus on data accuracy and reliability was reflected in rigorous processes like Lagrange interpolation for handling mid-sequence missing data and the application of quartiles and interquartile range (IQR) for outlier removal. This scrupulous approach ensured the robustness of our sample, which comprised 887 companies from 2016 to 2022. Moreover, we measured key variables such as Return on Assets (ROA), patents, innovation culture, R&D spend, among others. The gathered descriptive statistics offered an expansive view of these variables, laying the groundwork for our analysis of the interplay between innovation culture and organizational performance.

The critical role of this stage in theory generation lies in its ability to concretely anchor theoretical constructs with real-world data. It effectively narrows the gap between theoretical assumptions and empirical reality, bolstering the validity and practicality of our theoretical model. By quantitatively assessing key variables, we facilitate a refined analysis of the relationship between innovation culture and performance, which is essential for understanding the dynamics in China high-tech sector. The integration of this stage with preceding steps underscores a seamless transition from theoretical model construction to empirical validation, marking a notable progression in the realm of management theory generation.

Module 2: Empirical analysis and validation

This module encompasses both the empirical analysis and the validation of the theoretical model, employing advanced techniques such as GWO-SVM. It systematically quantifies the relationships between innovation culture and key performance metrics, rigorously testing the theoretical assumptions against real-world data. This comprehensive empirical approach not only validates the theoretical constructs but also provides concrete evidence for the effectiveness of the proposed theory, ensuring its robustness and applicability across different organizational contexts.

Stage 4. Discovering associations using SVM analysis.

This stage of is pivotal for delineating the intricate relationships among key factors like innovation culture, leverage, technician staff ratio, and corporate age, in relation to firm performance. This stage marks a transition from theoretical framework to empirical analysis. In this stage, we initially employ GWO to fine-tune the parameters of the SVM algorithm. The most significant advantage and role of GWO is its ability to reveal complex relationships among multiple factors, providing a comprehensive and nuanced perspective for a deeper understanding of the issue (Dinesh and Svn, 2024; Mirjalili et al., 2014).

This step is critical to ensure the precision of our model in capturing the complex dynamics within our dataset. SVM then meticulously explores the individual and combined influence of each factor on firm performance. By altering pairs of variables within the SVM framework, we gain insights into the complex interaction effects.

Our methodology, leveraging advanced ML techniques, offers a nuanced examination of relationships and interaction effects, surpassing the capabilities of traditional statistical methods. GWO, traditionally associated with leadership hierarchy simulations, is instrumental here in optimizing SVM parameters. This illustrates the adaptability of ML algorithms in unveiling hidden variable relationships, thereby boosting our model's explanatory power. This detailed exploration strengthens the empirical foundation of our theoretical model, emphasizing the impact of specific factors in China high-tech manufacturing sector.

Stage 5. Empirical validation of SVM insights.

In the empirical validation phase of our study, we meticulously align our SVM model's theoretical insights with the empirical realities of China high-tech manufacturing sector. This crucial phase marks a shift from abstract theorization to concrete data examination, aiming to substantiate the SVM model's predictions with tangible evidence.

Our methodology employs advanced statistical techniques, including regression analysis and Sobel tests, to rigorously examine the interplay between key factors like R&D spend, innovation culture, and firm performance. This phase is instrumental in verifying or refuting hypotheses based on our theoretical model, providing a robust empirical foundation for our theoretical constructs. We utilize regression analyses to test the relationships among critical variables, and Sobel tests to explore potential mediating effects and intricate interactions between these variables.

This stage offers a comprehensive and rigorous approach compared to traditional hypothesis testing methods, unveiling deeper insights and more complex relationships within the data. It is pivotal in theory generation, linking theoretical assumptions with empirical data, thus enhancing model relevance and robustness. This phase not only validates the theoretical framework developed in earlier stages but also enriches it with empirical evidence, contributing significantly to the understanding of dynamics in China high-tech manufacturing industry.

Module 3: Theory refinement and practical Implications

The final module is devoted to refining the theoretical constructs using AI-driven tools such as GPT-4, enhancing the clarity and applicability of the proposed theory. This stage ensures that the developed model contributes to theoretical advancement while offering actionable insights into the dynamics of organizational performance across diverse contexts.

Stage 6 theoretical construct development.

Stage 6 capitalizes on the innovative potential of AI in the realm of theoretical development. This stage is characterized by the application of the GPT-4 language model to interpret and refine our proposed theory, employing various narrative styles to elucidate the theory's aspects. This stage underscores the role of generative AI tools in theoretical workshops, highlighting their proficiency in expressing theories through different communicative styles. It demonstrates a significant advancement in integrating AI into management theory generation, offering a comprehensive AI-driven solution that complements and extends the findings from ML algorithms.

The approach involves a detailed exploration of the management theory through diverse narrative perspectives. This method facilitates a deeper understanding and effective communication of the theory's essence, showcasing its multifaceted dimensions. It also emphasizes the importance of examining theoretical constructs through different narrative styles to enhance human comprehension and identify the most effective mode of theoretical expression. It not only exhibits the capabilities of AI in augmenting theoretical understanding but also reflects our commitment to a thorough and nuanced exploration of theoretical concepts in management studies. It represents a novel attempt in the field, providing valuable insights for businesses, especially in the high-

tech manufacturing sector, and guiding them through the complex landscape of resource optimization and innovation-driven growth.

In our study, we employ an innovative algorithm-driven approach to generate management theory, particularly focusing on the high-tech manufacturing sector. We have structured our methodology into a cohesive framework. The table below clearly delineates the three modules of our research, encompassing the establishment of the theoretical framework, empirical analysis and validation, and the refinement and practical application of theoretical constructs. Through this systematic approach, we demonstrate how advanced methods such as LDA, GWO-SVM, and GPT-4 can be effectively utilized within a single research context to advance and expand management theory. [Table 1](#) provides an overview of the key steps in our methodological process and highlights the added value of each approach, aiding in the understanding of how these methods contribute to the generation and advancement of management theory.

5. An illustrate example: Analysis of the factors influencing organizational performance

5.1. Building a theoretical model of organizational performance

In this paper, the LDA modeling algorithm is employed to build the conceptual model. Firstly, we searched for relevant previous studies and obtained the corpus. Then, we browse the acquired textual data and perform data pre-processing. The number of topics is determined by using the coherence score C_{UMass} . Subsequently, the LDA modeling method is applied to get the specific topics of the literatures. Finally, we identify conceptual models based on the topics discovered. The main steps are shown as follows:

Step 1. Corpus Compilation: Leveraging the Web of Science Core Collection, we systematically mined literature for articles encapsulating “culture”, “innovation”, and “performance” within their titles, abstracts, or keywords. This meticulous search yielded a corpus of abstracts distilled from the identified articles, establishing the primary textual dataset for our subsequent analysis.

Step 2. Data pre-processing. We initiated the process by tokenizing the textual content, breaking it down into individual words based on spacing. Then, As removing stopwords can increase the number of remaining terms with topic-specific semantic content, also the model fit and quality can be enhanced by deleting determiners, conjunctions, and prepositions ([Schofield et al., 2017](#)), we delete the above types of words from the documents. Meanwhile, we drop punctuation marks, numbers, line codes, and words that are less than 4 in length. Due to words with the same root usually grouped under the same topic and stemming of words is unnecessary and often runs the risk of reducing quality of resulting model ([Schofield et al., 2017](#)), only lemmatization rather than stemming is used to combine near-duplicate terms. Finally, the data pre-processing process is complete.

Step 3. Selecting the appropriate number of topics. Different choice of parameters (e.g., the number of topics Z) will generate various topic distributions. Accuracy indicators such as perplexity and coherence are often applied to measure the number of topics. however, some scholars are skeptical about the relationship between the performance of quantitative metrics and the semantic quality of the topic ([Chang et al., 2009](#), [DiMaggio et al., 2013](#)). As a result, we consider both the algorithmic accuracy principle and the semantic interpretability principle. In our study, coherence scores across a spectrum of 1 to 20 topics were meticulously calculated for LDA models. Notably, the models comprising 2 to 8 topics demonstrated considerably high coherence scores. This prompted a closer investigation of the LDA models within this topic range and a comparative analysis of their respective results. Ultimately, our investigation culminated in the selection of a five-topic model. This choice was grounded in its optimal balance between comprehensive coverage and precise specificity, its ability to distinctly delineate between categories, and its strategic avoidance of excessive

Table 1

Algorithm-driven theory generation process in high-tech manufacturing management.

Module	Stage	Description	Traditional Approach	Our Approach	Advantages of Our Approach
Module 1: Theoretical Framework Establishment	Stage 1: Forming Theoretical Scope	Establish a foundational theoretical framework based on organizational performance, innovation, and culture	Literature review and manual analysis to construct theoretical frameworks	Use of LDA modeling to analyze large corpus data and identify key themes	Provides an unbiased and comprehensive literature review, reveals nuanced themes, and lays a robust foundation for subsequent analysis
	Stage 2: Extraction of Innovation Culture	Identify and quantify the cultural aspects driving organizational innovation	Qualitative analysis of annual reports and management discussions	Application of BERT model to extract key innovation culture aspects from textual data	Offers detailed and quantifiable insights into innovation culture, reduces biases, and enhances understanding of cultural dynamics
	Stage 3: Quantifying Theoretical Scopes	Measure key variables (e.g., ROA, patents) to ground theoretical constructs in empirical data	Use of predefined variables based on theoretical assumptions	Empirical measurement and data processing using advanced techniques such as Lagrange interpolation and IQR for outlier removal	Anchors theoretical models in real-world data, enhances validity, and supports more refined analysis of innovation culture's impact
Module 2: Empirical Analysis and Validation	Stage 4: Discovering Associations Using SVM Analysis	Explore complex relationships among variables such as innovation culture, leverage, and firm performance	Linear regression models to explore relationships among variables	Use of SVM analysis fine-tuned with GWO to examine nonlinear interactions and multi-variable effects	Provides a nuanced understanding of complex relationships, enhances model precision, and uncovers hidden variable interactions
	Stage 5: Empirical Validation of SVM Insights	Validate theoretical insights with empirical data to ensure robustness and applicability	Basic hypothesis testing using linear models	Use of advanced statistical methods (regression analysis, Sobel tests) to validate SVM predictions	Ensures robustness of theoretical constructs, links model predictions to real-world data, and strengthens empirical evidence for theory
Module 3: Theory Refinement and Practical Implications	Stage 6: Theoretical Construct Development	Refine theoretical models and ensure practical applicability across diverse contexts	Refinement based on manual review and expert judgment	Utilize GPT-4 to refine theory, employing diverse narrative styles for clearer articulation	Enhances theoretical clarity and adaptability, leverages AI for theory refinement, and integrates findings from ML algorithms into a coherent theoretical framework

fragmentation, thereby ensuring a focused and cohesive analysis. Fig. 2 displays the coherence scores for different numbers of topics, allowing us to observe which topic numbers had higher coherence scores. Subsequently, we compared the interpretability of themes across different topic counts and found that the five-topic model was the most suitable. Fig. 3 presents the word cloud results for the five-topic model, providing an intuitive visual representation of the themes.

Step 4. Generating Theoretical Scopes. Upon establishing the optimal number of topics through LDA modeling, we extracted pivotal themes encapsulating culture, performance, innovation, management, and sustainability. These themes collectively shed light on the scholarly discourse's focal points. Key observations include the synergy between organizational culture and performance, the pivotal role of innovation

in fortifying firm capabilities, and the influence of environmental sustainability on corporate practices. A notable aspect is the integration of green strategies and sustainable practices within organizational frameworks, alongside the exploration of knowledge management's role in bolstering organizational effectiveness. The outcomes of our LDA model, delineated in Table 2, unravel the complex interconnections and thematic richness of our study. This analysis prompts us to closely examine the interplay of technology, innovation, and firm performance in the evolving landscape of modern business dynamics.

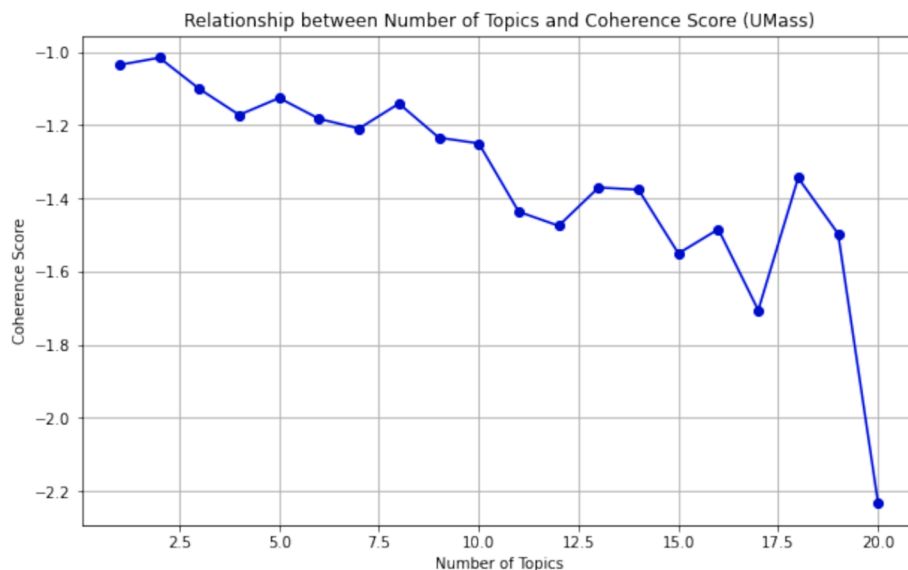
**Fig. 2.** Changes of coherence score with different number of topics.



Fig. 3. Topic word cloud for five topics.

Table 2
The raw topics of the LDA model and the results of corresponding scopes.

Theme Title	Curated Key Terms	Theme Description
#1 Organizational Dynamics and Performance	culture, performance, innovation, knowledge, organizational, management, technology, firm	This theme encapsulates the interplay of culture, innovation, and technology within firms, focusing on how these elements impact firm performance and management strategies.
#2 Green Practices and Innovation in Business	innovation, performance, organizational, firm, practice, green, culture, organization	This topic explores the relationship between environmental consciousness, innovative practices, and organizational performance, highlighting the role of sustainable strategies in the corporate sector.
#3 The Synergy of Knowledge and Organizational Management	innovation, organizational, performance, management, organization, firm, knowledge, value	Concentrating on how knowledge management, innovation, and organizational structures contribute to firm performance and business processes
#4 Cultural Influence and Innovation in Corporate Strategy	performance, culture, innovation, firm, organizational, business, value, relationship, service	This theme examines the integration of culture and innovative practices within business models, analyzing their impact on organizational performance and market positioning.
#5 Sustainability and Knowledge Management in Business	knowledge, culture, management, innovation, organizational, performance, business, cultural	Focusing on the role of cultural factors, sustainability practices, and knowledge management in enhancing organizational performance and fostering business development.

5.2. Enhancement of the innovation culture lexicon and quantitative assessment in corporate reports

In this section, we advanced the lexicon pertaining to “innovation culture” through a BERT-based model, strategically trained on the “Management Discussion and Analysis” sections extracted from the annual reports of China high-tech manufacturing firms (2016–2022). This textual choice was instrumental, as it provided a fertile ground for rich insights into corporate philosophies and strategic orientations, thus effectively encapsulating the core elements of organizational culture. Inspired by (Li et al., 2020), the processes commenced with key “seed

words” resonant of innovation culture including “innovation”, “creativity”, “passion”, and “efficiency”. The model was then leveraged to identify words with a high degree of semantic congruence, resulting in an expanded lexicon that embodies various facets of organizational innovation. The seed words and the results of the expanded lexicon are presented in [Table 3](#). Subsequent to lexicon expansion, our analysis shifted to quantifying its manifestation, delving into the frequencies of these terms across the complete textual corpus of the annual reports. This segment of the study unveils an innovative method for decoding and quantifying the subtleties of innovation culture within the realm of high-tech manufacturing. Harnessing cutting-edge language models, tailored to specific corporate contexts, enabled us to distill a nuanced lexicon that mirrors the prevailing culture of innovation in these entities.

5.3. Data measurement of key factors from Chinese High-Tech manufacturing enterprises

In our quest to delve into the interplay between innovation culture and performance in Chinese manufacturing enterprises, we focused on high-tech manufacturing companies listed in China as our research subjects. Given China's status as the world's second-largest economy, its manufacturing sector, particularly the high-tech segment, plays a pivotal role in the global industrial landscape, characterized by unique developmental patterns and managerial practices. The experiences of Chinese firms in innovation and technological advancement offer a wealth of empirical research cases for management studies. The strong emphasis placed by the Chinese government on innovation and high-tech industries renders these enterprises ideal candidates for exploring the nexus between corporate innovation culture and performance. Furthermore, the stringent disclosure requirements for listed companies guarantee more accurate and transparent data, providing a high-quality data source for our research.

For our study, we selected high-tech manufacturing companies listed in China, established before 2016, excluding those under PT or ST. We gathered our data from the Tianyancha website, Wind database, Patsnap database, and corporate annual reports. To ensure the accuracy and reliability of our data, we adopted the following steps:

a. Middle-value missing data treatment: We utilized Lagrange interpolation to fill mid-sequence missing values in our dataset, a technique apt for time-series data, ensuring data continuity and smoothness.

b. End-value missing data treatment: We applied specific methods for missing values at the ends of sequences. If both ends of a subsequence were missing, we filled them with the average of the adjacent values; for a missing end, we used the first non-missing value.

Table 3
The results of seed words and the expanded lexicon.

Seed Words	Translated English Expanded Lexicon
Innovation	First-Class, Continuous Progress, Interactive, Promote, Create First-
Innovate	Class, Creative Power, Shaping, Creativity, Creative Thinking,
Innovative	Innovation, Innovative Strength, Innovativeness, Construct, Create,
Creativity	Creativity, Creative, Creatively, Creative Ability, Spirit of Creation,
Creative	Creator, Brave, Resolute, Excellence, Eclectic, Inventive Creation,
Create	Era of Transformation, Sculpting, Multidimensional, Perfection,
Passion	Real-Time, Pioneering, Courageous Heart, Superior Performance,
Passionate	Imagination, Talent, Fighting Spirit, Newly Created, New Idea, Easy
Efficiency	to Control, Intelligent Manufacturing Power, More Superior,
Efficient	Upgrading, Maximum, Efficiency, Modularization, Flexibilization,
Excellence	Love, Devotion to Duty, Originality, Vision, Wisdom, Science Park,
Pride	Technology Era, Breakthrough, Determined, Creating, Fabricate,
	Self-Respect, Self-Enhancement, Self-Improvement, Pride, Utmost
	Glory, Devotion to Country, Heroic, Advanced Development,
	Surpass, Transcend, Change of Concept, Revolutionary, Innovation,
	Disruptive, First Creation, Pride, Arrogance, High Yield, High
	Performance, High Efficiency, High Precision, High Integration

c. Exclusion of companies with excessive data gaps: To maintain the integrity and representativeness of our research sample, we removed companies with key financial indicators missing for more than two years.

d. Calculation of annual averages and outlier removal: We first computed the annual average values for key financial indicators like ROA, R&D Spend, and patent grants. Subsequently, we identified and eliminated outliers in these average value columns using IQR, thereby minimizing the impact of data anomalies on our analysis.

Through these stringent selection and processing steps, our research ultimately encompassed 887 high-tech manufacturing companies listed in China from 2016 to 2022, providing a solid data foundation for an in-depth analysis of the relationship between corporate innovation culture and performance.

Table 4 presents the descriptive statistics of key variables from a sample of 6209 observations in the context of high-tech manufacturing firms in China. The variables include ROA, Patents, Innovation culture, R&D spend, Leverage, Revenue, Firm age, Employees, Technical staff, and Technical staff ratio. The table outlines each variable's average, variability, and range.

5.4. SVM insights on key drivers of firm performance

In this research, we meticulously evaluate the influence of various factors, including innovation culture, leverage, technician staff ratio, and company age, on firm performance. Our methodology begins with the utilization of the GWO to ascertain the most effective parameters for the SVM algorithm. This critical step ensures the precision of our model, enabling it to accurately reflect the intricate dynamics present in our data. Upon fine-tuning the SVM with these parameters, our attention shifts to the detailed examination of each factor's individual effect on firm performance. To address the challenge of ML algorithms being perceived as “black boxes”, we simulate the SVM's prediction trends with changes in specific variables, revealing the method's internal workings. After 34 iterations, the GWO algorithm converges, yielding a coefficient of determination $R^2 = 0.7695$, outperforming traditional linear models. We accomplish the analysis of individual effects by normalizing our variables and incrementally modifying the value of each factor (adjusting from 0 to 1 in 0.01 increments), while maintaining other variables at their median. This procedural adjustment permits us to capture variations in projected firm performance, elucidating the distinct influence of each factor. Moreover, to probe into the interaction effects, we scrutinize the combined influence of these factors on corporate outcomes. This involves concurrently tweaking the values of two factors within our SVM framework and tracking the consequent changes in the predicted outcomes. Such a multifaceted approach allows us to reveal the more intricate and layered interactions that might be overlooked in simpler, single-variable analyses. This comprehensive method offers deeper insights into the myriad factors that shape firm performance.

Upon garnering the results for the marginal association, Fig. 4 delineates a complex relationship between corporate innovation culture

and ROA. An initial escalation in innovation culture correlates with increased ROA, signifying a positive impact on firm performance. However, when innovation culture exceeds a specific threshold, a downturn in ROA suggests potential overinvestment in innovation or a disconnect with market requirements. The analysis also reveals that an enhanced tech staff ratio is tied to improved firm performance, with ROA significantly rising after surpassing a certain level, indicating the vital contribution of tech staff to profitability.

Moreover, Fig. 4 indicates that as firms age, there's a slight increase in ROA, hinting at the benefits of maturity and market establishment. Yet, with continued aging, there's a substantial upsurge in ROA, reflecting the significant effects of accumulated experience and established market presence. In contrast, the leverage-ROA relationship portrays an inverted U-shaped trajectory. An initial increase in leverage suggests that strategic borrowing can bolster a firm's investment returns, but as leverage heightens, ROA peaks and subsequently plummets, revealing the detrimental impacts of excessive financial risk.

Furthermore, Fig. 4 provides insight into the interactive dynamics between R&D spending and ROA within the context of varying leverage levels. With no leverage, an upward trend in ROA is linked to increased R&D spending, endorsing the intrinsic benefits of R&D initiatives. As leverage rises to moderate levels, the growth in ROA becomes more tempered, albeit from a higher plateau, which infers that moderate leveraging could potentially facilitate R&D endeavors. When leverage is high, although there is a positive association between R&D spending and ROA, the ROA performance at higher levels of R&D spending is not as strong as when leverage is moderate or low. This suggests that while high leverage may enhance the impact of R&D on firm performance, the overall ROA does not reach the levels seen under moderate or low leverage. This interaction between leverage and R&D spending invites further investigation into the delicate equilibrium of financial strategies that can maximize the effectiveness of R&D on firm performance.

5.5. Empirical validation of SVM insights in China High-Tech manufacturing sector

In this study, we bridge the gap between theoretical insights derived from our SVM model and the empirical realities of China high-tech manufacturing sector. The SVM model, with its focus on innovation culture, leverage, and other critical factors, lays the groundwork for an empirical validation. This phase marks a crucial shift from theoretical abstraction to concrete data examination, striving to corroborate the SVM's theoretical predictions with real-world evidence.

Our study delves into the dynamics between innovation efforts and firm performance within China high-tech manufacturing sector. The regression analysis of the impact of R&D on ROA and innovation culture, presented in Table 5, showing R&D spend significantly boosts ROA ($\beta = 8.73\text{e-}09$, $p < 0.01$), highlighting the payoff of innovation activities. Model (2) confirms Hypothesis 2, showing a robust link between innovation culture and enhanced performance ($\beta = 7.02\text{e-}08$, $p < 0.01$), suggesting that nurturing an innovative culture can drive success. For Hypothesis 3, the evidence from Model (3) affirms the mediating role of

Table 4
Descriptive Statistics of Key Variables.

Variable	Obs	Mean	Std. Dev.	Min	Max
ROA	6209	3.854886	7.025999	-59.2716	54.0959
Patents	6209	13.80287	17.21782	0	342
Innovation Culture	6209	47.11	30.08494	0	384
R&D Spend	6209	1.02E + 08	1.07E + 08	-5.98E-08	1.42E + 09
Leverage	6209	36.57934	17.38979	1.4268	122.9411
Revenue	6209	2.86E + 09	4.49E + 09	7.10E + 07	1.05E + 11
Firm Age	6209	20.22322	5.593549	7	64
Employees	6209	2637.356	3374.217	51	43,228
Tech Staff	6209	444.2099	557.1226	8	10,672
Tech Staff Ratio	6209	19.85241	12.22914	0.67	82.07

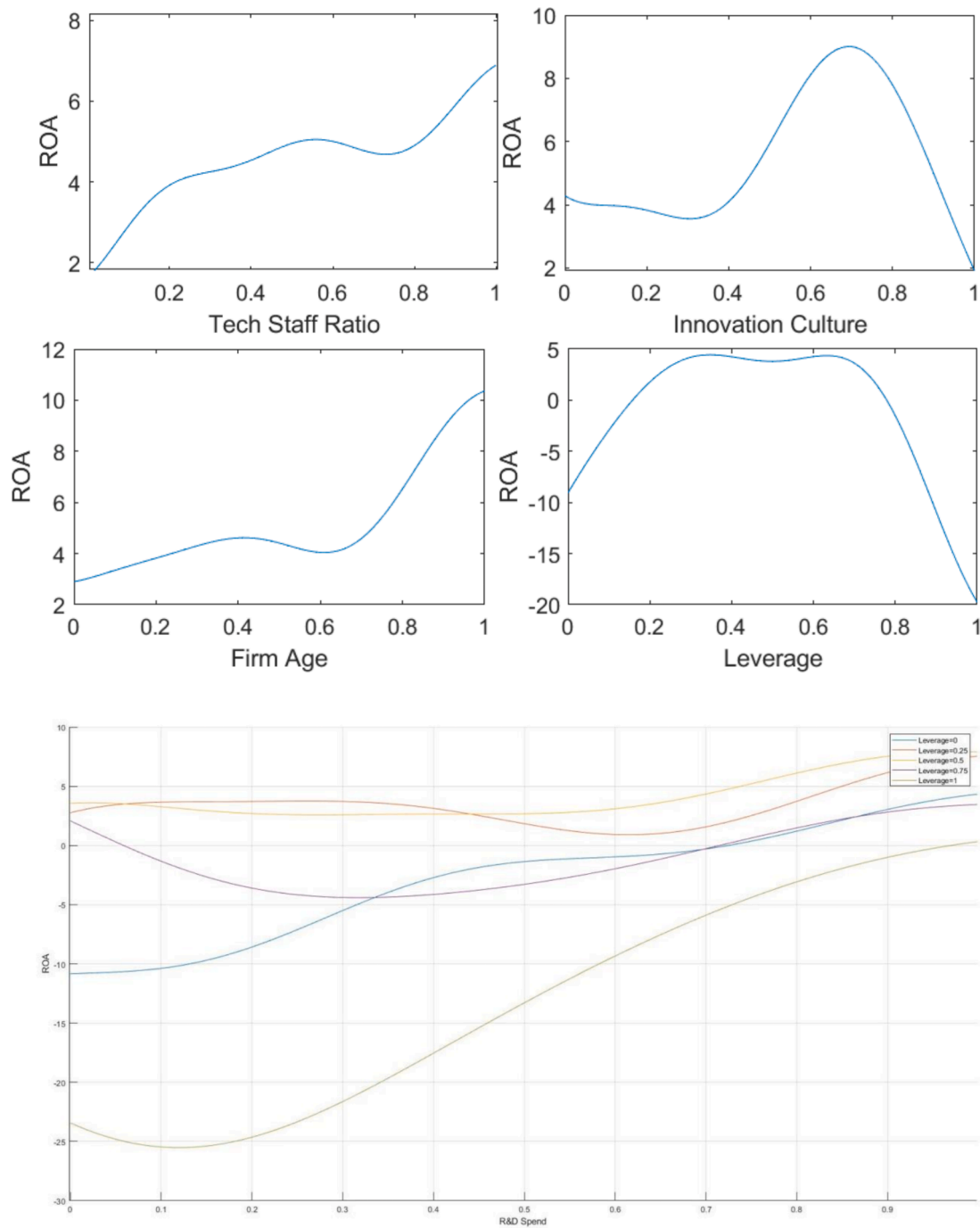


Fig. 4. The outcome of the relationship between the variables.

innovation culture in strengthening the impact of R&D on performance ($\beta = 0.0048567$, $p < 0.05$). The Sobel test underlines this mediation, with a significant indirect effect (coef = 3.4×10^{-10} , $p < 0.1$), suggesting innovation culture significantly channels the influence of R&D on firm performance. Lastly, Model (4) provides backing for Hypothesis 4, where a higher leverage is found to negatively adjust the R&D-ROA relationship ($\beta = -2.30 \times 10^{-10}$, $p < 0.01$), indicating that while some debt can be beneficial, too much may hinder the positive effects of R&D.

Patents as an independent variable strengthens mediation analysis. The results of the robustness test for the mediation effect are presented in Table 6. Model (5) demonstrates that patents significantly boost ROA ($\beta = 0.012282$, $p < 0.05$), suggesting patents contribute positively to firm performance. Model (6) aligns with Hypothesis 2, indicating that

patents positively affect innovation culture ($\beta = 0.1106858$, $p < 0.01$). Model (7) reinforces Hypothesis 3, highlighting innovation culture's mediating role between patents and ROA ($\beta = 0.00861$, $p < 0.05$). The Sobel test confirms the mediation's stability, with a significant indirect effect (coef = 0.000953 , $p < 0.01$), showing that innovation culture significantly channels the impact of patents on ROA.

The empirical data closely mirrors the SVM model's predictions, underlining the significant roles played by R&D spending and innovation culture in shaping firm performance. This harmony between theoretical modeling and empirical validation strengthens our conclusions. The robust evidence from regression analysis and the Sobel test corroborate the mediating effects, illustrating the complex interrelations that drive performance in China's advanced manufacturing sector. Thus,

Table 5

Regression Analysis of R&D Impact on ROA and Innovation Culture.

Model	(1)	(2)	(3)	(4)
VARIABLES	ROA	Innovation Culture	ROA	ROA
R&D Spend	8.73e-09*** (1.03e-09)	7.02e-08*** (4.69e-09)	8.39e-09*** (1.05e-09)	1.38e-08*** (3.22e-09)
Innovation Culture			0.0048567* (0.0027847)	
Leverage * R&D Spend				-2.30e-10*** (6.01e-11)
Leverage	-0.1663286*** (0.004911)	0.0417113* (0.0223899)	-0.1665312*** (0.0049115)	-0.182*** (0.0108)
Revenue	2.23e-10*** (2.71e-11)	-1.50e-10 (1.24e-10)	2.23e-10*** (2.71e-11)	5.24e-10*** (0)
Firm Age	-0.0503038*** (0.0146695)	-0.0624291 (0.0668805)	-0.0500006*** (0.0146681)	-0.281*** (0.0419)
Employees	0.00005 (0.0000412)	-0.0012832*** (0.0001879)	0.0000563 (0.0000414)	-0.000238** (0.000107)
Tech Staff	-0.0005687** (0.0002255)	0.0046971*** (0.0010279)	-0.0005916*** (0.0002258)	-0.000109 (0.000521)
Constant	9.545709*** (0.3434275)	41.38582*** (1.565736)	9.34471*** (0.3621955)	14.92*** (0.794)
Observations	6209	6209	6209	6209
Number of ID	887	887	887	887
R-squared	0.1694	0.0583	0.1698	0.120

Standard errors in parentheses.

*** p < 0.01, ** p < 0.05, * p < 0.1.

Table 6

Robustness Test of Mediation Effect in Patent Influence on ROA.

Model	(5)	(6)	(7)
VARIABLES	ROA	Innovation Culture	ROA
Patents	0.012282** (0.0048254)	0.1106858*** (0.0222316)	0.011329*** (0.0048316)
Innovation Culture			0.00861** (0.0027541)
Leverage	-0.162804*** (0.0049205)	0.0692615*** (0.0226701)	-0.1634004*** (0.0049208)
Revenue	3.25e-10*** (2.45e-11)	6.73e-10*** (1.13e-10)	3.19e-10*** (2.45e-11)
Firm Age	-0.0453967*** (0.0147351)	-0.0239112 (0.0678882)	-0.0451908*** (0.0147249)
Employees	0.0000145 (0.0000413)	-0.0015718*** (0.0001902)	0.0000281 (0.0000415)
Tech Staff	-9.92e-06 (0.0002158)	0.009148*** (0.000994)	-0.0000887 (0.0002171)
Constant	9.594462*** (0.3467822)	41.68726*** (1.597707)	9.235533*** (0.3650616)
Observations	6209	6209	6209
Number of ID	887	887	887
R-squared	0.1606	0.0282	0.1619

Standard errors in parentheses.

*** p < 0.01, ** p < 0.05, * p < 0.1.

our comprehensive approach not only affirms our initial hypotheses but also highlights the intricate dynamics influencing corporate success.

5.6. AI-enhanced theoretical Exploration: Unveiling the dynamic Resource-Culture Synergy theory

Building upon the innovative ideas presented in (Jarvenpaa and Klein, 2024), their study explores the application of generative AI tools as collaborative participants in theoretical workshops. These AI tools are adept at articulating theories in diverse communicative styles, determining which styles, such as propositional, narrative, or typological, most precisely convey theoretical concepts. This research introduces the “Dynamic Resource-Culture Synergy Theory”, an advancement of the traditional RBV, enriched by the insights from our empirical findings. Inspired by the potentials of AI in theoretical innovation, we employ the GPT-4 language model to interpret this newly proposed theory through various narrative lenses. This approach marks a significant step in

integrating AI into the realm of management theory generation. It presents a comprehensive AI-driven solution that complements the robust findings derived from ML algorithms, aiding in the culmination of theory development. Moreover, by examining the theory through different narrative styles, we integrate human comprehension to identify the most effective mode of theoretical expression. This endeavor is not only a testament to the capabilities of AI in enhancing theoretical understanding but also reflects our commitment to deeply comprehend and effectively communicate the essence of the “Dynamic Resource-Culture Synergy Theory”. Here we provide an in-depth exploration of the theory, articulated through diverse narrative styles, showcasing its multifaceted dimensions.

1 Propositional Style: The dynamic Resource-Culture Synergy Theory, introduces an innovative perspective for explaining and predicting the performance of high-tech manufacturing enterprises. This theory emphasizes that the success of a company depends not only on the scarce and unique resources mentioned in traditional RBV but also relies more on the dynamic synergistic effect between these resources and innovation culture. In this theory, resources are no longer viewed as isolated entities. Their value and efficacy depend on how they are dynamically utilized and integrated within a specific organizational culture and structure. For instance, resources such as R&D investments and patent activities, driven by a strong innovation culture, can be transformed more effectively into competitive advantages for the company. The dynamic integration and utilization of resources become key to enhancing business performance. Additionally, the theory highlights the significant role of organizational structure in the process of resource utilization. Organizational structures, such as leverage ratios and the proportion of technical staff, can affect the efficiency and effectiveness of resource allocation, thereby impacting business performance. The dynamic Resource-Culture Synergy theory stresses that understanding the enhancement of business performance requires going beyond the analysis of single resources, delving into the complex interactions between resources, organizational culture, and structure.

2 Story Narration: In a bustling high-tech manufacturing enterprise in China, CEO Zhang Hua faces a challenge. Despite significant investments in R&D and patents, his company's performance has not reached expectations. Upon deep analysis, he realizes that the issue lies not in the quantity or quality of resources but in how effectively these resources are utilized. Zhang Hua begins to transform his management

strategy, focusing on cultivating the company's innovation culture. He encourages innovative thinking and knowledge sharing among employees, breaking down the barriers between departments, and promoting cross-departmental collaboration. He also adjusts the company's organizational structure to better support innovation projects and ensure that resources can flow effectively within the company. Over time, these changes begin to show effects. The company's R&D team becomes more efficient, and innovation projects make significant progress. The number and quality of patents increase markedly, and the company's market position is thereby strengthened. Zhang Hua realizes that these achievements are not solely due to the resources themselves but more so due to how these resources are effectively integrated and utilized under the push of the company's innovation culture.

3 Typological Style: According to the dynamic Resource-Culture Synergy Theory, high-tech manufacturing enterprises can be classified into different types, reflecting their distinct characteristics in resource management, innovation culture, and organizational structure. For example, one type of enterprise may invest heavily in R&D and patents, but without corresponding innovation culture and appropriate organizational structure support, these resources may not be fully transformed into market competitive advantages. Another type of enterprise may have a strong innovation culture and, even with fewer resources, can achieve significant performance improvements through effective resource integration and utilization. This typological analysis offers a new perspective to understand and explain how different enterprises achieve success in the market based on their resource combination, organizational culture, and structure. For instance, some enterprises with abundant technical resources might not translate these into actual market advantages due to a lack of effective cultural support and organizational mechanisms. In contrast, enterprises with fewer resources could achieve higher market performance through a strong innovation culture and flexible organizational structure, effectively integrating and utilizing existing resources.

Based on the diverse narrative interpretations, we propose the concept of the "Dynamic Resource-Culture Synergy Theory" which presents an extension that goes beyond the traditional scope of the RBV. It posits that the value and effectiveness of resources in an organization are fundamentally connected to a dynamic interplay with the organization innovation culture. This theory re-conceptualizes resources, not as isolated or static elements, but as dynamic components whose potential is realized through interactions within the organizational culture and structure. From a conceptual standpoint, the "Dynamic Resource-Culture Synergy Theory" implies that when resources like R&D investments and patents are synergistically aligned with a strong innovation culture, they can be more effectively harnessed to secure competitive advantages. This dynamic integration and utilization of resources are critical for enhancing business performance and ensuring sustainable growth. Moreover, the theory highlights the critical role of organizational structure in enabling efficient resource utilization. Elements such as leverage and the ratio of technical staff play a crucial role in influencing how resources are distributed and used, significantly impacting the overall performance of the business. Furthermore, "Dynamic Resource-Culture Synergy Theory" offers a holistic framework for comprehending the complex interplay between resources, innovation culture, and organizational structure. It advocates a comprehensive approach to resource management, emphasizing a strategy that acknowledges the dynamic and interactive nature of resources within the organizational environment. This theory provides valuable insights for businesses, particularly in the high-tech manufacturing sector, guiding them through the intricate landscape of resource optimization and innovation-led growth.

6. Discussion

6.1. Key findings and normative analysis

Organizational performance is central to organizational theory, with organizational culture recognized as a key determinant of performance growth (Oyemomi et al., 2019). However, research in this area varies, as seen in studies like Arabeche et al. (2022), which explored the mediating role of organizational culture in innovation and performance in SMEs. Divergent perspectives, backgrounds, and limited empirical evidence have led to inconsistent conclusions and the absence of a unified, systematic viewpoint. This study introduces a data-driven research methodology, including normative analysis, to explore the interactive relationships among organizational performance, organizational culture, and innovation. The methodology unfolds across six stages: forming the theoretical scope, extracting innovation culture, quantifying theoretical constructs, discovering associations, empirically validating the insights, and refining the theoretical constructs. Using panel data from 887 high-tech manufacturing companies listed in China from 2016 to 2022, and drawing on RBV and Dynamic Capability theory, this approach examines the relationship between innovation and firm performance in the context of organizational culture, accounting for factors such as leverage, firm size, and the proportion of technical personnel. By leveraging advanced tools—LDA for topic modeling, SVM for relational analysis, and GPT-4 for theory refinement—the study offers key insights into the factors shaping organizational performance in the digital era.

The empirical analysis provides significant insights into the interactions between R&D expenditure, innovation culture, and firm performance, particularly within the high-tech manufacturing sector. The results confirm that innovation serves as a key driver of firm performance, with a positive and significant effect on ROA. This finding supports the theoretical expectation that firms with greater R&D spending tend to achieve higher profitability, underscoring the critical role of innovation activities in sustaining competitive advantage.

Moreover, the study reveals that innovation culture plays a crucial role, both as a direct contributor to performance and as a mediator in the relationship between innovation and firm outcomes. Firms that cultivate a strong innovation-oriented culture are more effective at converting their R&D investments into tangible performance gains, highlighting the strategic importance of fostering an innovation-driven culture to maximize the returns on R&D activities.

The analysis further demonstrates that leverage moderates the relationship between innovation and firm performance. While moderate levels of leverage amplify the positive effects of innovation investment on ROA, excessive leverage diminishes these benefits, introducing financial risk that can undermine the impact of innovation efforts. This suggests that firms should carefully manage their debt levels to strike an optimal balance that enhances innovation performance without overburdening the organization financially.

6.2. Theoretical contributions

The Dynamic Resource-Culture Synergy Theory offers significant advantages over traditional theories by integrating the RBV with organizational culture theory, providing a more dynamic and holistic perspective on the formation of competitive advantage. This theory transcends the limitations of static models by emphasizing the dynamic mediating role of culture in resource integration and innovation processes, thereby enhancing the explanatory power regarding how firms maintain competitiveness in rapidly changing environments. Furthermore, grounded in data-driven empirical research, this theory possesses high predictive capability and broad applicability, making it effective across different organizational and industry contexts and fostering further research exploration. Here are the key theoretical contributions of this study:

1 Extension of the RBV

This study extends the application of the RBV, which posits that a firm's competitive advantage stems from its unique and inimitable resources and capabilities. Through empirical analysis, our research reveals that organizational innovation culture—characterized by traits such as creativity, passion, and innovative—serves as an intangible resource that mediates the relationship between innovation activities and competitive advantage. This cultural environment stimulates employees' innovative behaviors, thereby enhancing organizational performance and competitiveness. This finding addresses the existing literature's relatively broad discussion of cultural impacts and provides a clearer theoretical understanding of the specific roles of cultural characteristics in the innovation process.

2 Enrichment of organizational culture Theory

This study also enriches organizational culture theory by specifically examining the nuances of innovative organizational culture. Unlike the broader discussions in existing literature, our research delves into how specific cultural characteristics mediate the transformation of innovation into enhanced organizational performance. This nuanced exploration provides more precise insights into the role of cultural types in innovation and firm performance, filling gaps in the literature's general discussions on cultural impacts (Naveed et al., 2022), and building upon insights from the research by Hogan and Coote (2014). This repositioning not only deepens the understanding of Schein's model by emphasizing the critical role of cultural artifacts in transforming innovative culture into organizational performance but also enhances its applicability in explaining complex organizational phenomena and innovation processes.

3 Utilization of the competing values framework (CVF)

Additionally, this study aligns with the findings of Naranjo-Valencia et al. (2011) and further supported by Naranjo-Valencia et al. (2016). This paper employs the CVF to refine the mediating role of culture in the transformation of innovation into competitive advantage. By analyzing the “flexibility vs. control” dimension of the CVF (Büschgens et al., 2013a), the study emphasizes how cultural traits characterized by flexibility more effectively support the process of turning innovation into competitive advantage. These cultural traits facilitate a faster response to market changes, identification of new opportunities, and implementation of innovations, thereby enriching existing organizational culture theories, particularly in understanding how effective connections are formed between innovation activities and competitive advantages.

6.3. Methodological Innovation

Methodologically, this study introduces an innovative quantitative measurement approach by utilizing natural language processing techniques and ML algorithms (such as BERT models and LDA topic modeling) to identify and quantify cultural artifacts in Chinese annual reports. This approach addresses the limitations of traditional qualitative methods by incorporating quantitative analysis tools into cultural research, thereby making the study of cultural artifacts more systematic and precise. The data-driven approach enhances the applicability of this theory across different organizational and industry contexts and aids in uncovering the micro-level mechanisms of culture-resource interactions, thereby deepening the understanding of how firms sustain competitive advantage in dynamic environments.

In contrast to previous studies that primarily focused on data analysis and theoretical assumptions (Nguyen et al., 2021), the LDA model, compared to traditional manual review, screening, and summarization methods, extracts topics from large-scale texts, reduces researcher biases, and strikes a balance between model complexity and interpretability (Zhang et al., 2021). The application of the LDA model provides a robust foundation for empirical analysis, ensuring a close link between theory and practice. Furthermore, our study employs a SVM analysis as a control group instead of traditional econometric methods (Haffar et al., 2022, Scaliza et al., 2021, Nguyen et al., 2021), highlighting the

significant advantages of ML algorithms, such as SVM, in handling large-scale, complex data and uncovering nonlinear relationships. We also optimize the theory using GPT-4, enabling in-depth exploration and understanding of the theory's essence and structure from different narrative perspectives, and facilitating a deep integration of research in this field with academic frontiers, providing a valuable theoretical foundation for enterprise resource optimization and innovation-driven development.

In the study of organizational culture, we innovatively employ advanced BERT models. Compared to static questionnaires or scales (Xie et al., 2023), this method can understand and process complex relationships in context, handle language ambiguity to some extent, and better capture the dynamic and diverse nature of organizational culture. It facilitates a detailed exploration of micro and macro features of organizational culture, providing a multi-dimensional cultural analysis. Importantly, it lowers the knowledge threshold for research participants, as the model can learn language representations without explicit annotations or reliance on prior knowledge, making it adaptable to different fields and cultural differences.

6.4. Practical implications

Based on the research findings, the following recommendations are proposed for the future development of enterprises and providing guidance for making practical business decisions:

This study emphasizes the strategic importance of fostering an innovation-driven culture within organizations. For practitioners, this entails creating an environment that encourages creativity, passion, and continuous innovation. By prioritizing cultural initiatives aligned with market dynamics, organizations can enhance their adaptability and leverage innovation to achieve a sustainable competitive advantage. The Dynamic Resource-Culture Synergy Theory underscores that organizational culture is a dynamic asset, critical for optimizing resource integration and capability development. Therefore, organizations should continuously align their cultural practices with evolving strategic objectives and environmental demands. Innovation must be recognized as a key driver of long-term growth, with clear and specific innovation goals that are regularly assessed for alignment with the company's overall strategy. Mechanisms should be established to monitor the health of the innovation culture and manage risks, particularly when innovation efforts exceed certain thresholds. Leaders should balance innovation with other essential business elements, such as operational management, customer relations, employee well-being, and quality control, to ensure comprehensive and sustainable growth.

Moreover, conduct regular assessments of the impact of internal culture, capital structure, and other critical factors on the organization. Adjust strategies based on market changes and enterprise development. Flexible strategic adjustments assist the enterprise in adapting to the continuously evolving business environment. Pay particular attention to the following aspects: maintain an optimal ratio of technical personnel to support innovation and technological development; cautiously select leverage ratios and assess their impact on R&D investment to maximize return on investment, avoiding excessive leverage that may lead to a slowdown in investment returns; closely monitor the influence of company age, recognizing that over time, market positioning may improve. Be cautious of potential over-aging that could result in a decline in competitiveness, and adopt flexible strategies to maintain market competitiveness.

While the research in this paper is based on the manufacturing sector in China, the conclusions also involve some universal principles such as balancing innovation culture and prudently managing capital structure. These principles are applicable to various types of enterprises as they are generally considered good practices in corporate management, guiding decision-making in different environments. However, it is inevitable that opportunities and challenges faced by different countries, ethnicities, and regions vary. The development goals and national cultures also

exhibit differences, leading to potentially distinct perceptions of innovation culture. Furthermore, the practicality of the conclusions may change over time due to variations in data and time spans utilized in this study. When applying these conclusions, customized adjustments based on specific industry and regional contexts are recommended.

7. Conclusion and future research directions

In our comprehensive study, we rigorously investigated the complexities of organizational performance, culture, and innovation within the high-tech manufacturing sector, utilizing an innovative, multi-staged approach. The employment of LDA modeling was pivotal in developing a nuanced theoretical framework, effectively synthesizing a vast corpus of literature to illuminate the synergies between these crucial elements. Subsequently, through the advanced BERT model, we enhanced the lexicon of innovation culture, gaining deeper insights into its multifaceted nature within corporate settings. Our empirical analysis, drawing on data from 887 Chinese high-tech manufacturing firms, sheds light on key determinants of organizational performance. The GWO-SVM technique allowed us to unravel complex relationships among various factors, including innovation culture and leverage. This exploration revealed that while an initial increase in innovation culture positively impacts performance metrics such as ROA, excessive emphasis on innovation culture might lead to performance hindrances, suggesting a need for balanced innovation strategies.

Future research should examine how various organizational contexts—such as industry sector, geographic location, and firm size—impact the effectiveness of innovation culture as a mediator between resources and performance. Comparative and longitudinal studies across diverse settings can provide insights into the dynamic nature of culture and its evolving influence on innovation. While this study focuses on the Chinese high-tech manufacturing context, expanding to other industries and geographies, alongside incorporating different theoretical frameworks and methodologies, could offer a more comprehensive and generalizable understanding of the culture-innovation-performance nexus.

CRedit authorship contribution statement

Wanqing Li: Writing – review & editing, Writing – original draft, Visualization, Supervision, Software, Methodology, Formal analysis, Data curation, Conceptualization. **Jiang Yu:** Writing – review & editing, Writing – original draft, Validation, Supervision, Resources, Funding acquisition, Formal analysis, Conceptualization. **Feng Chen:** Writing – review & editing, Writing – original draft, Supervision, Methodology, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

References

- Arabeche, Z., Soudani, A., Brahmi, M., Aldieri, L., Vinci, C. P., & Abdelli, M. E. A. (2022). Entrepreneurial Orientation, Organizational Culture and Business Performance in SMEs: Evidence from Emerging Economy. *Sustainability*, 14(9), 5160.
- Arora, A., Gittelman, M., Kaplan, S., Lynch, J., Mitchell, W., & Siggelkow, N. (2016). Question-based innovations in strategy research methods. *Strategic Management Journal*, 37(1), 3–9.
- Bamel, U., Kumar, S., Lim, W. M., Bamel, N., & Meyer, N. (2022). Managing the dark side of digitalization in the future of work: A fuzzy TISM approach. *Journal of Innovation & Knowledge*, 7(4), Article 100275.
- Barney, J. B. (1986). "Organizational culture: Can it be a source of sustained competitive advantage?" *The Academy of Management Review*, 11(3), 656–665. <https://doi.org/10.2307/258317>
- Barney, J. B. (1991). Firm Resources and Sustained Competitive Advantage. *Journal of Management*, 17, 120–199.
- Bendak, S., Shikhli, A. M., & Abdel-Razek, R. H. (2020). How changing organizational culture can enhance innovation: Development of the innovative culture enhancement framework. *Cogent Business & Management*, 7.
- Boubakri, N., Chkir, I., Saadi, S., & Zhu, H. (2021). "Does national culture affect corporate innovation? International evidence." *Journal of Corporate Finance*.
- Büschgens, T., Bausch, A., & Balkin, D. B. (2013a). "Organizing for radical innovation — A multi-level behavioral approach." *The Journal of High Technology Management Research*, 24, 138–152.
- Cao, L. (2022). A new age of AI: Features and futures. *IEEE Intelligent Systems*, 37(1), 25–37.
- Carmeli, A., & Gefen, D. (2005). "The relationship between work commitment models and employee withdrawal intentions." *Journal of Managerial Psychology*, 20, 63–86.
- Chang, J., S. Gerrish, C. Wang, J. Boyd-Graber and D. Blei (2009). "Reading tea leaves: How humans interpret topic models." *Advances in neural information processing systems* 22.
- Cefis, E., & Marsili, O. (2006). "Survivor: The Role of Innovation in Firms' Survival. *Journal of Catalysis*".
- Charles, V., et al. (2022). Why data analytics is an art. *Significance (Oxford, England)*, 19 (6), 42–45.
- Chatterjee, S., Chaudhuri, R., Mikalef, P., & Sarpong, D. (2023). "Coopetition in the platform economy from ethical and firm performance perspectives." *Journal of Business Research*.
- Choudhury, P., Allen, R. T., & Endres, M. G. (2021). Machine learning for pattern discovery in management research. *Strategic Management Journal*, 42(1), 30–57.
- Crown, W. H. (2019). "Real-World Evidence, Causal Inference, and Machine Learning". *The journal of the International Society for Pharmacoeconomics and Outcomes Research*, 22(5), 587–592.
- Daft, R.L. (1983). "Organization Theory and Design".
- Dani, M. V., & Gandhi, A. V. (2021). "Understanding the drivers of innovation in an organization: A literature review. *International Journal of Innovation Science*".
- Davis, J. P., Eisenhardt, K. M., & Bingham, C. B. (2007). Developing theory through simulation methods. *Academy of Management Review*, 32(2), 480–499.
- Denison, D. R. (1990). *Corporate culture and organizational effectiveness*. John Wiley & Sons.
- Deshpande, P. S., et al. (2019). Predictive and prescriptive analytics in big-data era. *Security and data storage aspect in cloud computing*, 71–81.
- DiMaggio, P., Nag, M., & Blei, D. (2013). Exploiting affinities between topic modeling and the sociological perspective on culture: Application to newspaper coverage of US government arts funding. *Poetics*, 41(6), 570–606.
- Dinesh, K., & Svn, S. K. (2024). GWO-SMSLO: Grey wolf optimization based clustering with secured modified Sea Lion optimization routing algorithm in wireless sensor networks. *Peer-to-Peer Networking and Applications*, 1–27.
- Eisenhardt, K. M., & Graebner, M. E. (2007). Theory building from cases: Opportunities and challenges. *Academy of Management Journal*, 50(1), 25–32.
- Ghoshal, S. (2005). Bad management theories are destroying good management practices. *Academy of Management Learning & Education*, 4(1), 75–91.
- Griffith, D. A., & Rubera, G. (2014). A Cross-Cultural Investigation of New Product Strategies for Technological and Design Innovations. *Journal of International Marketing*, 22, 20–25.
- Grover, V. (2020). Do We Need to Understand the World to Know It? Knowledge in a Big Data World. *Journal of Global Information Technology Management*, 23(1), 1–4.
- Gunday, G., Ulusoy, G., Kilic, K., & Alpkın, L. (2011). Effects of innovation types on firm performance. *International Journal of production economics*, 133(2), 662–676.
- Haffar, M., Al-Karaghoul, W., Djebarni, R., Al-Hyari, K. A., Gbadamosi, G., Oster, F., Alaya, A., & Ahmed, A. (2022). Organizational culture and affective commitment to e-learning' changes during COVID-19 pandemic: The underlying effects of readiness for change. *Journal of Business Research*, 155, 113396.
- Hannigan, T.R., Haans, R.F., Vakili, K., Tchalhan, H., Glaser, V.L., Wang, M.S., Kaplan, S., & Jennings, P.D. (2019). "Topic Modeling in Management Research: Rendering New Theory from Textual Data." *Academy of Management Annals*.
- Hogan, S. J., & Coote, L. V. (2014). Organizational culture, innovation, and performance: A test of Schein's model. *Journal of Business Research*, 67(8), 1609–1621. <https://doi.org/10.1016/j.jbusres.2013.09.007>
- Homburg, C., & Pflesser, C. (2000). "A Multiple-Layer Model of Market-Oriented Organizational Culture: Measurement Issues and Performance Outcomes." *Journal of Marketing Research*, 37, 449–462.
- Huang, J., Henfridsson, O., & Liu, M. J. (2022). Extending Digital Ventures Through Templating. *Information Systems Research*, 33(1), 285–310.
- Jarvenpaa, S., & Klein, S. (2024). *New Frontiers in Information Systems Theorizing: Human-AI Collaboration*. Syst. J. Assoc. Inf.

- Jiang, Z., Ma, G., & Zhu, W. (2022). Research on the impact of digital finance on the innovation performance of enterprises. *European Journal of Innovation Management*.
- Kafetzopoulos, D., & Psomas, E. L. (2015). The impact of innovation capability on the performance of manufacturing companies. *Journal of Manufacturing Technology Management*, 26, 104–130.
- Kar, A. K., & Dwivedi, Y. K. (2020). Theory building with big data-driven research - Moving away from the "What" towards the "Why". *International Journal of Information Management*, 54.
- Khan, H. U., Ali, M., Olya, H., Zulqarnain, M. N., & Khan, Z. R. (2018). Transformational leadership, corporate social responsibility, organizational innovation, and organizational performance: Symmetrical and asymmetrical analytical approaches. *Corporate Social Responsibility and Environmental Management*.
- Kumar, V., Ramachandran, D., & Kumar, B. (2020). Influence of new-age technologies on marketing: A research agenda. *Journal of Business Research*.
- Lepeniotti, K., et al. (2020). Prescriptive analytics: Literature review and research challenges. *International Journal of Information Management*, 50, 57–70.
- Leung, A.-K.-Y., Maddux, W. W., Galinsky, A. D., & Chiu, C.-Y. (2008). Multicultural experience enhances creativity: The when and how. *American Psychologist*, 63(3), 169–181. <https://doi.org/10.1037/0003-066X.63.3.169>
- Li, K., Mai, F., Shen, R., & Yan, X. (2020). *Measuring Corporate Culture Using Machine Learning*. SPGMI: Capital IQ Data (Topic).
- Liu, J., Zhou, H., Chen, F., & Yu, J. (2022). The coevolution of innovation ecosystems and the strategic growth paths of knowledge-intensive enterprises: The case of China's integrated circuit design industry. *Journal of Business Research*, 144, 428–439.
- Lu, Q. L., & Chesbrough, H. (2022). Measuring open innovation practices through topic modelling: Revisiting their impact on firm financial performance. *Technovation*, 114.
- Mariani, M. M., & Wamba, S. F. (2020). Exploring how consumer goods companies innovate in the digital age: The role of big data analytics companies. *Journal of Business Research*, 121, 338–352.
- Martínez, J. M. G., Carracedo, P., Comas, D. G., & Siemens, C. H. (2022). An analysis of the blockchain and COVID-19 research landscape using a bibliometric study. *Sustainable Technology and Entrepreneurship*, 1(1), Article 100006. <https://doi.org/10.1016/j.stae.2022.100006>
- Martins, E. C., & Terblanche, F. (2003). Building organisational culture that stimulates creativity and innovation. *European Journal of Innovation Management*, 6, 64–74.
- Menter, M., Göcke, L., Zeeb, C., & Clauss, T. (2023). "Disentangling the complex longitudinal relationships between business model innovation and firm performance." *Journal of Business Research*.
- Miotto, A. P., & Parente, J. (2015). Antecedents and consequences of household financial management in Brazilian lower-middle-class. *Rae-revista De Administracao De Empresas*, 55, 50–64.
- Mirjalili, S., Mirjalili, S. M., & Lewis, A. (2014). Grey wolf optimizer. *Advances in engineering software*, 69, 46–61.
- Miric, M., Jia, N., & Huang, K. G. (2023). Using supervised machine learning for large-scale classification in management research: The case for identifying artificial intelligence patents. *Strategic Management Journal*, 44(2), 491–519.
- Murphy, C. B., Klotz, A. C., & Kreiner, G. E. (2017). Blue skies and black boxes: The promise (and practice) of grounded theory in human resource management research. *Human Resource Management Review*, 27, 291–305.
- Naghsbandi, M. M., & Tabche, I. (2018). The interplay of leadership, absorptive capacity, and organizational learning culture in open innovation: Testing a moderated mediation model. *Technological Forecasting and Social Change*.
- Nauhaus, S., Luger, J., & Raisch, S. (2021). Strategic Decision Making in the Digital Age: Expert Sentiment and Corporate Capital Allocation. *Journal of Management Studies*, 58(7), 1933–1961.
- Nguyen, T., Viet Ngo, L., & Paramita, W. (2021). Turning lurkers into innovation agents: An interactionist perspective of self-determinant theory. *Journal of Business Research*.
- Nyagadza, B. (2022). Sustainable digital transformation for ambidextrous digital firms: A systematic literature review and future research directions. *Sustainable Technology and Entrepreneurship*, 100020. <https://doi.org/10.1016/j.stae.2022.100020>
- Oyemomi, O., Liu, S., Neaga, I., Chen, H., & Nakpodia, F. (2019). How cultural impact on knowledge sharing contributes to organizational performance: Using the fsQCA approach. *Journal of Business Research*.
- Ozdemir, S., Fernandez, C., de Arroyabe, J., Sena, V., & Gupta, S. (2023). Stakeholder diversity and collaborative innovation: Integrating the resource-based view with stakeholder theory. *Journal of Business Research*.
- Rajapathirana, R. P., & Hui, Y. (2017). Relationship between innovation capability, innovation type, and firm performance. *Journal of Innovation & Knowledge*, 3, 44–55.
- Rollinson, D. J. (2002). "Organisational Behaviour and Analysis: An Integrated Approach".
- Saeed, A., Alnori, F., & Yaqoob, G. (2023). Corporate social responsibility, industry concentration, and firm performance: Evidence from emerging Asian economies. *Research in International Business and Finance*.
- Scaliza, J. A., Jugend, D., Chiappetta Jabbour, C. J., Latan, H., Armellini, F., Twigg, D. R., & Andrade, D. F. (2021). Relationships among organizational culture, open innovation, innovative ecosystems, and performance of firms: Evidence from an emerging economy context. *Journal of Business Research*.
- Schein, E. H. (1992). *Organizational culture and leadership (2nd ed)*. San Francisco: Jossey-Bass.
- Schofield, A., Magnusson, M., Thompson, L., & Mimmo, D. (2017). Understanding text pre-processing for latent Dirichlet allocation. *Proceedings of the 15th conference of the European chapter of the Association for Computational Linguistics*.
- Shepherd, D. A., & Suddaby, R. (2017). Theory building: A review and integration. *Journal of Management*, 43(1), 59–86.
- Shrestha, Y. R., He, V. F., Puranam, P., & von Krogh, G. (2021). Algorithm Supported Induction for Building Theory: How Can We Use Prediction Models to Theorize? *Organization Science*, 32(3), 856–880.
- Skare, M., & Soriano, D. R. (2021). How globalisation is changing digital technology adoption: An international perspective. *Journal of Innovation & Knowledge*, 6(4), 222–233. <https://doi.org/10.1016/j.jik.2021.04.00>
- Škerlavaj, M., Song, J. H., & Lee, Y. (2010). Organizational Learning Culture, Innovative Culture and Innovations in South Korean Firms. *ERN: Culture & Leadership (Topic)*.
- Subramanian, A., & Nilakanta, S. (1996). Organisational innovativeness: Exploring the relationship between organisational determinants of innovation, types of innovations, and measures of organisational performance. *Omega*, 24(6), 631–647.
- Svevig, P. (2021). A Meta-theoretical framework for theory building in project management. *International Journal of Project Management*, 39(8), 849–872.
- Tan, B. S. (2019). In search of the link between organizational culture and performance: A review from the conclusion validity perspective. *Leadership & Organization Development Journal*, 40(3), 356–368.
- Upmeyer zu Belzen, A., Engelschalt, P., & Krüger, D. (2021). "Modeling as Scientific Reasoning—The Role of Abductive Reasoning for Modeling Competence." *Education Sciences*.
- Usai, A., Fiano, F., Petruzzelli, A. M., Paoloni, P., Briamonte, M. F., & Orlando, B. (2021). "Unveiling the impact of the adoption of digital technologies on firms' innovation performance." *Journal of Business Research*, 133, 327–336.
- Wamba-Taguidje, S. L., Fosso Wamba, S., Kala Kamdjoug, J. R., & Tchatchouang Wanko, C. E. (2020). Influence of artificial intelligence (AI) on firm performance: The business value of AI-based transformation projects. *Business Process Management Journal*, 26(7), 1893–1924.
- Walker, R. M., 2004. "Innovation and Organizational Performance: Evidence and a Research Agenda." *Advanced Institute of Management Research Working Paper*.
- Weick, K. E., & Quinn, R. E. (1999). Organizational change and development. *Annual review of psychology*, 50, 361–386.
- Weinzimmer, L. G., Esken, C. A., Michel, E. J., McDowell, W. C., & Mahto, R. V. (2023). The differential impact of strategic aggressiveness on firm performance: The role of firm size. *Journal of Business Research*.
- Xie, Q., Gao, Y., Xia, N., Zhang, S., & Tao, G. (2023). Coopetition and organizational performance outcomes: A meta-analysis of the main and moderator effects. *Journal of Business Research*.
- Xu, Y. J., Liu, X., Cao, X., Huang, C. P., Liu, E. K., Qian, S., Liu, X. C., Wu, Y. J., Dong, F. L., Qiu, C. W., Qiu, J. J., Hua, K. Q., Su, W. T., Wu, J., Xu, H. Y., Han, Y., Fu, C. G., Yin, Z. G., Liu, M., Roepman, R., Dietmann, S., Virta, M., Kengara, F., Zhang, Z., Zhang, L. F., Zhao, T. L., Dai, J., Yang, J. L., Lan, L., Luo, M., Liu, Z. F., An, T., Zhang, B., He, X., Cong, S., Liu, X. H., Zhang, W., Lewis, J. P., Tiedje, J. M., Wang, Q., An, Z. L., Wang, F., Zhang, L. B., Huang, T., Lu, C., Cai, Z. P., Wang, F., & Zhang, J. B. (2021). Artificial intelligence: A powerful paradigm for scientific research. *Innovation*, 2(4).
- Yu, J., Wen, Y., Jin, J., & Zhang, Y. (2019). Towards a service-dominant platform for public value co-creation in a smart city: Evidence from two metropolitan cities in China. *Technological Forecasting and Social Change*, 142, 168–182.
- Zhang, Z., Zhu, H., Zhou, Z., & Zou, K. (2022). How does innovation matter for sustainable performance? Evidence from small and medium-sized enterprises. *Journal of Business Research*.
- Jacques, E. (1952). *The changing culture of a factory*. New York: Dryden.
- Mohammadi, M., Samizadeh, M., Pouya, S., & Arabahmadi, R. (2023). Examining the mediating effect of knowledge management on the relationship between organizational culture and organizational performance. *Journal of Soft Computing and Decision Analytics*, 1(1), 63–79.
- Naveed, R. T., Alhaidan, H., Al Halbusi, H., & Al-Swidi, A. K. (2022). Do organizations really evolve? The critical link between organizational culture and organizational innovation toward organizational effectiveness: Pivotal role of organizational resistance. *Journal of Innovation & Knowledge*, 7(2), Article 100178.
- Naranjo-Valencia, J. C., Jiménez-Jiménez, D., & Sanz-Valle, R. (2016). Studying the links between organizational culture, innovation, and performance in Spanish companies. *Revista latinoamericana de psicología*, 48(1), 30–41.
- Naranjo-Valencia, J. C., Jiménez-Jiménez, D., & Sanz-Valle, R. (2011). Innovation or imitation? The role of organizational culture. *Management decision*, 49(1), 55–72.
- Büschgens, T., Bausch, A., & Balkin, D. B. (2013b). Organizational culture and innovation: A meta-analytic review. *Journal of product innovation management*, 30(4), 763–781.

Wanqing Li received the B.S. degree in Business Administration from Capital University of Economics and Business in 2019 and the M.S. degree in Big data management from Southwest University of Finance and Economics in 2022. She is currently working toward the Ph.D. degree in Public Administration at the Institutes of Science and Development, Chinese Academy of Sciences. Her research interests include AI-driven innovation and corporate performance under digitalization.

Jiang Yu is a Professor at the Institutes of Science and Development, Chinese Academy of Sciences. His main research interests include digital innovation, innovation policy and competitive strategy in emerging countries. He has worked as Royal Society International Fellow in the Department of Engineering of Cambridge University. His book, *China's Highway of Information and Communication Technology*, was published by Palgrave Macmillan.

Feng Chen is a research associate at the Institutes of Science and Development, Chinese Academy of Sciences, Beijing. Her work focuses on the management of technological innovation and policy. Her current research interests include technological catching up,

industrial upgrading under digitalization and S&T policies in the contexts of large emerging countries.